

ALBERTO BOMBARDELLI

Postodoctoral researcher in Computer Science

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CURRENT ROLE & RESEARCH WORK

Program-specific Postdoctoral Researcher in Computer Science at Kyoto University, working on the ZT-IoT CREST project under the supervision of Prof. Atsushi Igarashi.
My research focuses on the formal verification of zero-trust IoT components specified in the Rabbit language, using the protocol verification tools Tamarin and ProVerif.
Previously, I received my Ph.D. in Computer Science from the University of Trento, supported by a doctoral scholarship funded by Fondazione Bruno Kessler, under the supervision of Prof. Stefano Tonetta.
My Ph.D. research focused on **Symbolic Model Checking** for trace properties and hyperproperties, with particular emphasis on **compositional reasoning** and **SMT-based verification**.

WORK EXPERIENCE

Program-specific Postodoctoral Researcher

Graduate School of Informatics, Kyoto University

Jan 2026 – ongoing — Kyoto, Japan

Worked on the Zero Trust IoT CREST project focusing on ProVerif and Tamarin protocol verifiers.
Supervisor: Prof. Atsushi Igarashi

Software Developer

Fondazione Bruno Kessler

2017 – 2021 — Trento, Italy

Part time C Software Developer on the formal verification tools OCRA, nuXmv, NuSMV and HyCOMP.
Supervisor: Stefano Tonetta

EDUCATION

International Graduate Visiting Student

IMDEA Software Institute

Jan-May 2024 — Madrid, Spain

I worked with Prof. César Sanchéz on hyperproperties.

PhD in Computer Science

University of Trento

2021-2025 — Trento, Italy

Supervisor: Stefano Tonetta

Master Degree in Computer Science

University La Sapienza of Rome

2019-2021 — Rome, Italy

Final Score: 110/110 cum laude

Bachelor Degree in Computer Science

University of Trento

2016-2019 — Trento, Italy

Final Score: 98/110

TECHNICAL SKILLS

Python | C/C++ | LaTeX | C# | Docker/Singularity | Yacc/BISON | ANTLR4 | Godot | Javascript

PROJECTS & TOOLS

Rabbit

Kyoto University

- Tool for verification of zt-iot systems.
- Contribution: Conceptualized translation from Rabbit language to ProVerif.

COMPASTA (ESA)

Fondazione Bruno Kessler

- Combine Early V&V tool with Architectural design and deployment tool
- Contribution: Defining the semantics of the hierarchy and the component scheduling.

OCRA

Fondazione Bruno Kessler

- Contract-based design tool for Requirements analysis of discrete and hybrid system
- Hierarchical Component-based System with contract expressed via temporal logic
- Contribution: maintenance, parametrized and timed mode, asynchronous composition.

nuXmv

Fondazione Bruno Kessler

- Tool for symbolic model checking of infinite-state Symbolic Transition System
- Contribution: bug fixing, integration of research work in the tool

PUBLICATIONS (PEER-REVIEWED)

LPAR26: "Induction and K-Induction for Hyperproperties"
Bombardelli, Sanchez, and Tonetta

LMCS26: "Asynchronous Composition of LTL Properties over Infinite and Finite Traces"
Bombardelli and Tonetta

SPIN25: "(Asynchronous) Temporal Logics for Hyperproperties on Finite Traces"
Bombardelli, Bozzelli, et al.

CAES24: "COMPASTA = COMPASS + TASTE"
Bombardelli, Bonizzi, Bozzano, Cavada, Cimatti, Griggio, Nicolodi, et al.

JPK60: "Another Look at LTL Modulo Theory over Finite and Infinite Traces"
Bombardelli, Cimatti, Griggio, et al.

FSTTCS24: "Unifying Asynchronous Logics for Hyperproperties"
Bombardelli, Bozzelli, et al.

iFM23: "Symbolic Model Checking of Relative Safety LTL Properties"
Bombardelli, Cimatti, Tonetta, et al.

DATE23: "Metric Temporal Logic with Resettable Skewed Clocks"
Bombardelli and Tonetta

ADA23: "COMPASTA: Integrating COMPASS Functionality into TASTE"
Bombardelli, Bonizzi, Bozzano, Cavada, Cimatti, Griggio, Nazaria, et al.

NFM23: "Reasoning with Metric Temporal Logic and Resettable Skewed Clocks"
Bombardelli and Tonetta

NFM22: "Asynchronous Composition of Local Interface LTL Properties"
Bombardelli and Tonetta

IMBSA22: "COMPASTA: Extending TASTE with Formal Design and Verification Functionality"
Bombardelli, Bozzano, et al.

OTHER EXPERIENCES

Summer School
Marktoberdorf Summer School on Safety and Security through Formal Verification
Marktoberdorf, Germany Aug. 2023

Summer School
EuroProofNet Summer School Verification Technology,Systems & Applications
Saarbrucken, Germany Sep. 2022

Subreviewer
CAV22, TACAS23-24, FMCAD23, NFM23, FM24, CSL25,CAV26

Artifact Evaluation Committee
TACAS25,FM24,TACAS26,CAV26,VMCAI27

Program Committee
OVERLAY25

Journal Reviewers
FAC, TOSEM

LANGUAGES

Italiano (Native)
English (C1 self-assessed)
Español (A2 self-assessed)
日本語 (N4 self-assessed)

